WMO report documents spiralling weather and climate impacts

The clear signs of human-induced climate change reached new heights in 2024, with some of the consequences being irreversible over hundreds if not thousands of years, according to a new report from the World Meteorological Organization (WMO), which also underlined the massive economic and social upheavals from extreme weather.

Key messages

- Key climate change indicators again reach record levels
- Long-term warming (averaged over decades) remains below 1.5°C
- Sea-level rise and ocean warming irreversible for hundreds of years
- Record greenhouse gas concentrations combined with El Niño and other factors to drive 2024 record heat
- Early warnings and climate services are vital to protect communities and economies

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WMO's State of the Global Climate report confirmed that 2024 was likely the first calendar year to be more than 1.5°C above the pre-industrial era, with a global mean near-surface temperature of 1.55 ± 0.13 °C above the 1850-1900 average. This is the warmest year in the 175-year observational record.

WMO's flagship report showed that:

- Atmospheric concentration of carbon dioxide are at the highest levels in the last 800,000 years.
- Globally each of the past ten years were individually the ten warmest years on record.
- Each of the past eight years has set a new record for ocean heat content.
- The 18 lowest Arctic sea-ice extents on record were all in the past 18 years.
- The three lowest Antarctic ice extents were in the past three years.
- The largest three-year loss of glacier mass on record occurred in the past three

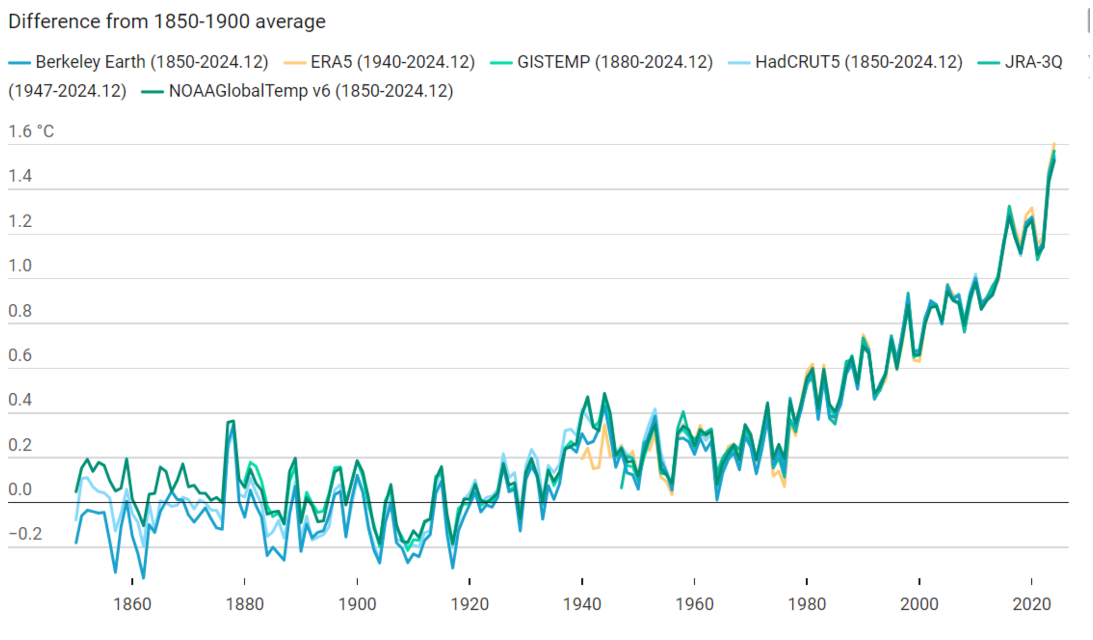
years.

- The rate of sea level rise has doubled since satellite measurements began.

“Our planet is issuing more distress signals -- but this report shows that limiting long-term global temperature rise to 1.5 degrees Celsius is still possible. Leaders must step up to make it happen -- seizing the benefits of cheap, clean renewables for their people and economies -- with new National climate plans due this year, ” said United Nations Secretary-General António Guterres.

“While a single year above 1.5 °C of warming does not indicate that the long-term temperature goals of the Paris Agreement are out of reach, it is a wake-up call that we are increasing the risks to our lives, economies and to the planet,” said WMO Secretary-General Celeste Saulo.

Global mean temperature 1850-2024



Annual global mean temperature anomalies relative to a pre-industrial (1850–1900) baseline shown from 1850 to 2024

Chart: WMO • Get the data • Embed • Download image • Created with Datawrapper

The report said that long-term global warming is currently estimated to be between 1.34 and 1.41 °C compared to the 1850-1900 baseline based on a range of methods – although it noted the uncertainty ranges in global temperature statistics.

A WMO team of international experts is examining this further in order to ensure consistent, reliable tracking of long-term global temperature changes to be aligned with the Intergovernmental Panel on Climate Change (IPCC).

Regardless of the methodology used, every fraction of a degree of warming matters and increases risks and costs to society.

The record global temperatures seen in 2023 and broken in 2024 were mainly due to the ongoing rise in greenhouse gas emissions, coupled with a shift from a cooling La Niña to warming El Niño event. Several other factors may have contributed to the unexpectedly unusual temperature jumps, including changes in the solar cycle, a massive volcanic eruption and a decrease in cooling aerosols, according to the report.

Temperatures are just a small part of a much bigger picture.

“Data for 2024 show that our oceans continued to warm, and sea levels continued to rise. The frozen parts of Earth’s surface, known as the cryosphere, are melting at an alarming rate: glaciers continue to retreat, and Antarctic sea ice reached its second-lowest extent ever recorded. Meanwhile, extreme weather continues to have devastating consequences around the world,” said Celeste Saulo.

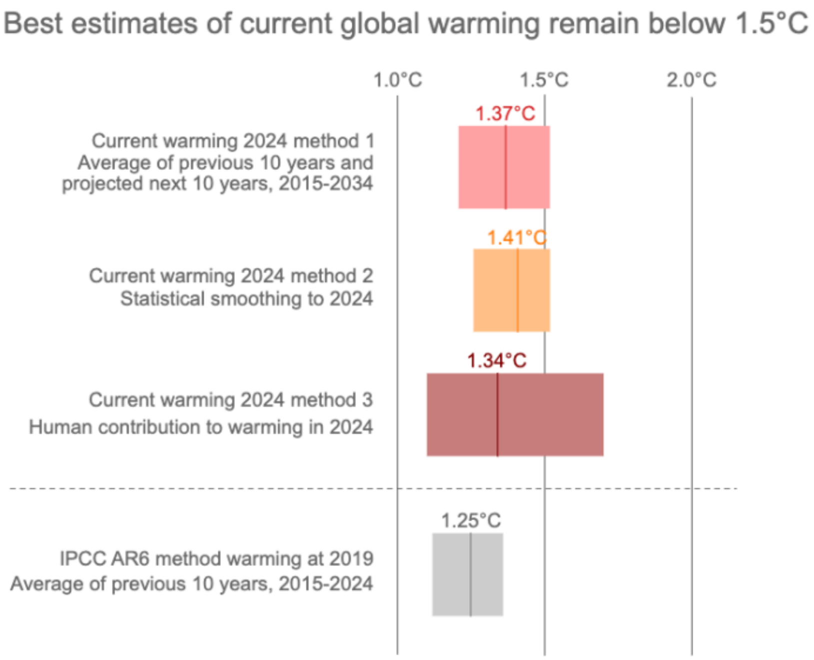
Tropical cyclones, floods, droughts, and other hazards in 2024 led to the highest number of new displacements recorded for the past 16 years, contributed to worsening food crises, and caused massive economic losses.

“In response, WMO and the global community are intensifying efforts to strengthen early warning systems and climate services to help decision-makers and society at large be more resilient to extreme weather and climate. We are making progress but need to go further and need to go faster. Only half of all countries worldwide have adequate early warning systems. This must change,” said Celeste Saulo.

Investment in weather, water and climate services is more important than ever to meet the challenges and build safer, more resilient communities, she stressed.

The report is based on scientific contributions from National Meteorological and Hydrological Services, WMO Regional Climate Centres, UN partners and dozens of experts. It includes sidebars on monitoring global temperature for the Paris Agreement and understanding the temperature anomalies in 2023 and 2024. It includes supplements on climate services and on extreme weather.

It is one of a suite of WMO scientific reports which seek to inform decision-making. It was published ahead of World Meteorological Day on 23 March, World Water Day on 22 March and World Glaciers Day on 21 March.



Three methods for establishing an up-to-date estimate of current global warming as of 2024, compared with the IPCC AR6 method, which uses averages over the previous 10 years and is representative of warming to 2019. The best estimate resulting from each method is shown as a dark vertical line, and the uncertainty range is shown by the shaded area.

Key Indicators

Atmospheric Carbon Dioxide

Atmospheric concentration of carbon dioxide, as well as methane and nitrous oxide, are at the highest levels in the last 800,000 years.

Carbon dioxide concentrations in 2023 (the last year for which consolidated global annual figures are available) were 420.0 ± 0.1 parts per million (ppm), 2.3 ppm more than 2022 and 151% of the pre-industrial level (in 1750). 420 ppm corresponds to 3,276 Gt – or 3.276 trillion tonnes of CO_2 in the atmosphere.

Real-time data from specific locations show that levels of these three main greenhouse gases continued to increase in 2024. Carbon dioxide remains in the atmosphere for generations, trapping heat.

Global Mean Near-surface Temperature

In addition to 2024 setting a new record, each of the past ten years, 2015-2024, were individually the ten warmest years on record.

The record temperature in 2024 was boosted by a strong El Niño which peaked at the start of the year. In every month between June 2023 and December 2024, monthly average global temperatures exceeded all monthly records prior to 2023.

Record levels of greenhouse gases were the primary driver, with the shift to El Niño playing a lesser role.

Ocean Heat Content

Around 90% of the energy trapped by greenhouse gases in the Earth system is stored in the ocean.

In 2024, ocean heat content reached its highest level in the 65-year observational record. Each of the past eight years has set a new record. The rate of ocean warming over the past two decades, 2005-2024, is more than twice that in the period 1960-2005.

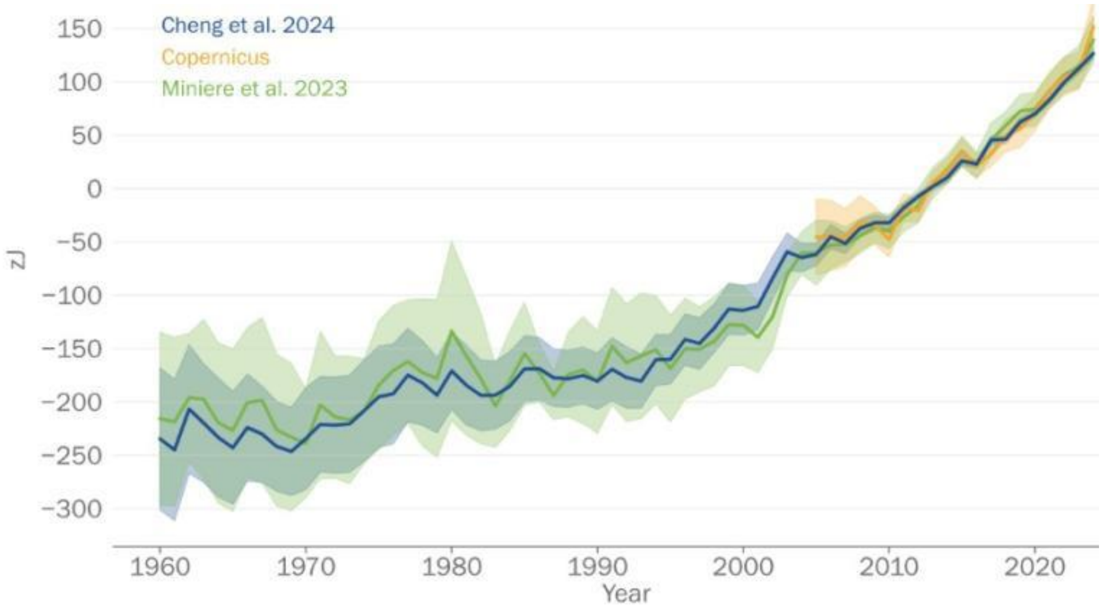
Ocean warming leads to degradation of marine ecosystems, biodiversity loss, and reduction of the ocean carbon sink. It fuels tropical storms and contributes to sea-level rise. It is irreversible on centennial to millennial time scales. Climate projections show that ocean warming will continue for at least the rest of the 21st century, even for low carbon emission scenarios.

Ocean Acidification

Acidification of the ocean surface is continuing, as shown by the steady decrease of global average ocean surface pH. The most intense regional decreases are in the Indian Ocean, the Southern Ocean, the eastern equatorial Pacific Ocean, the northern tropical Pacific, and some regions in the Atlantic Ocean.

The effects of ocean acidification on habitat area, biodiversity and ecosystems have already been clearly observed, and food production from shellfish aquaculture and fisheries has been hit as have coral reefs.

Projections show that ocean acidification will continue to increase in the 21st century, at rates dependent on future emissions. Changes in deep-ocean pH are irreversible on centennial to millennial time scales.



Annual global ocean heat content down to 2000 m depth for the period 1960–2024, in zettajoules (1021 J). The shaded area indicates the 2-sigma uncertainty range on each estimate.

Global Mean Sea Level

In 2024, global mean sea level was the highest since the start of the satellite record in 1993 and the rate of increase from 2015-2024 was double that from 1993–2002, increasing from 2.1 mm per year to 4.7 mm per year.

Sea level rise has cascading damaging impacts on coastal ecosystems and infrastructure, with further impacts from flooding and saltwater contamination of groundwater.

Glacier Mass Balance

The period 2022-2024 represents the most negative three-year glacier mass balance on record. Seven of the ten most negative mass balance years since 1950 have occurred since 2016.

Exceptionally negative mass balances were experienced in Norway, Sweden, Svalbard, and the tropical Andes.

Glacier retreat increases short-term hazards, harms economies and ecosystems and long-term water security.

Sea-ice Extent

The 18 lowest Arctic sea-ice minimum extents in the satellite record all occurred in the past 18 years. The annual minimum and maximum of Antarctic sea-ice extent were each the 2nd lowest in the observed record from 1979.

The minimum daily extent of sea-ice in the Arctic in 2024 was 4.28 million km2, the 7th lowest extent in the 46-year satellite record. In Antarctica, the minimum daily extent tied for the 2nd lowest minimum in the satellite era and marked the 3rd consecutive year that minimum Antarctic sea-ice extent dropped below 2 million km2. These are the three lowest Antarctic ice minima in the satellite record.